

Layers 5, 6 and 7 also called as network support layers.

▶ True

▶ False (P 42)

Internet with small "i" specifies the world wide Network the actual internet.

▶ True

▶ False (P 39)

Data link layer deals with mechanical and electrical specifications of transmission medium and interface

▶ True (P 44)

▶ False

Data synchronization is a function related with _____

▶ session layer (P 51)

▶ presentation layer

▶ transport

OSI model was developed prior to TCP/IP model.

▶ True (P 54)

▶ False

Fourier transform tells us that any digital signal can be decomposed into infinite number of periodic signals

▶ True

▶ False (P58)

A signal that repeats a pattern over a regular interval of time is called_____.

- ▶ **periodic signal (P58)**
- ▶ analog signal
- ▶ composite signal

Frequency band in which 99% of the total power resides is called_____

- ▶ **power bandwidth (P 67)**
- ▶ half power bandwidth
- ▶ 3dB bandwidth
- ▶ F.M.

In differential Manchester we require two transition or signal changes to represent binary_____

- ▶ zero
- ▶ **one (P76)**
- ▶ both zero and one

In Alternate mark inversion the term mark comes from_____

- ▶ **telegraphy (P77)**
- ▶ telephony
- ▶ digital telephony

A 56k modem can download at a rate of _____ Kbps and upload at a rate of _____ Kbps.

- ▶ 33.6: 33.6
- ▶ **33.6: 56.6**
- ▶ 56.6: 33.6

▶ 56.6: 56.6

The bit rate always equals the baud rate in which type of signal?

▶ **FSK (P 86)**

▶ QAM

▶ 4-PSK

▶ PSK

Asynchronous transmission is _____

▶ **slow (P 99)**

▶ costly

▶ non-effective

▶ fast

At the _____ layer, a DCE takes data generated by a DTE.

▶ **physical (P 101)**

▶ transport

▶ data link

▶ application

_____ requires more bandwidth.

▶ FSK

▶ **ASK (P 112)**

▶ PSK

▶ QAM

Which topology requires a central controller or hub?

- ▶ Mesh
- ▶ **Star (P 30)**
- ▶ Bus
- ▶ Ring

Unidirectional traffic movement is overcome by dual ring technology.

- ▶ **True (P33)**
- ▶ False

Which one best describes the given statement:

"To allow multiple users to share total capacity of a transmission medium"

- ▶ congestion control
- ▶ exchange management
- ▶ **multiplexing (P17)**

Which one is not among standard creation committee.

- ▶ **internet society and IETF (P 24)**
- ▶ ITU-T
- ▶ IEEE

Set of rules that govern communication is called

- ▶ OSI model
- ▶ **Protocol (P 8)**
- ▶ TCP/IP model
- ▶ Datagram

Frequency of a failure and network recovery time after a failure is measures of the _____ of a network.

- ▶ **Performance (P 14)**
- ▶ Reliability
- ▶ Security
- ▶ Feasibility

3dB is also called as power bandwidth.

- ▶ True
- ▶ **False (P66)**

ASK, PSK, FSK and QAM are examples of _____ modulation.

- ▶ digital-to-digital
- ▶ **digital-to-analog (P85)**
- ▶ analog-to-analog
- ▶ analog-to-digital

Packet switching is a connection oriented service.

- ▶ True
- ▶ **False (P38)**

Data synchronization is a function related with _____

- ▶ **session layer (P51)**
- ▶ presentation layer
- ▶ transport

Fourier transform tells us that any digital signal can be decomposed into infinite number of periodic signals

► True

► False (P 58)

Questions:

1. Describe different types of NRZ encoding scheme?

Answer:

In NRZ encoding the level of signal is either positive or negative. NRZ is classified into 2 types:

NRZ-L NRZ-I

NRZ-L:

Level of the signal depends on the type of bit it represents.

Positive voltage usually means the bit is a 1 and a negative voltage usually means the bit is a 0.

NRZ-I:

The inversion of the level represents a 1 bit. A bit 0 is represented by no change.

NRZ-I is superior to NRZ-L

2. Define Sampling Rate.

Answer: (page82)

The accuracy of any digital reproduction of an analog signal depends upon the o.
of samples taken

3. Write down the types of digital to analog modulation.

Answer: (PAGE 85)

Types of digital to analog modulation

ASK

FSK

PSK

QAM

4. What is periodic signal?

Answer: (page58)

A signal is called Periodic if it completes a pattern within a measurable time frame called a Period and then repeats that pattern over identical subsequent Periods

5. How bandwidth of a Modem is measured?

Answer: (page111)

Bandwidth of a modem is not measured in bytes, it is measured in bits.

Modem Speed –ASK (full-duplex)

Effective bandwidth of a telephone line is 2400 Hz, the maximum buad rate is also 2400.

Baud rate and bit rate are equal for ASK, so maximum bit rate is also 2400 bps.

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